hw1

1)

Let A be an nxn lower triangular nonsingular matrix with elements $a_{i,j}$.
Let B=A^{-1} and B not be lower triangular with elements $b_{i,j}$.
Know that AB=I and for $i \neq j$, $I_{i,j} = 0$ Let $B_{m,k}$ be the first nonzero entry above the main diagonal (m<k).

Then the prod of (row m of A) *(col k of B)*
*\$=a_{m,1}*b*{1,k} + a*{m,2}*b{2,k} + ... + a{m,n}*b*{n,k}\$, call this equation **1***
Know that for \$a{m,j}=0\$ where $j > m$ so **1** is equal to
$= a_{m,1} * b_{1,k} + a_{m,2} * b_{2,k} + \ldots + a_{m,m} * b_{m,k}$
Similarly $b_{1,k} = \ldots = b_{m-1,k} = 0$ so **1** is equal to
$= a_{m,m} * b_{m,k}$ and since $a_{m,m} \neq 0$ because A is nonsingular and lower triangular (diagonal must be nonzero because if there is a 0 in the main diagonal then the determinant is 0) and $b_{m,k} \neq 0$ because we picked it so
equation 1$\neq 0$

We have shown that $I_{m,k}$=(row m of A) * (col k of B) $\neq 0$ but $I_{m,k} = 0$ based on the definition of identity matrix so we have a contradiction

So B must be lower triangular

2

a)

$||x_1 + x_2||_2^2 = ||x_1||_2^2 + 2\langle x_1, x_2 \rangle + ||x_2||_2^2$ since x_1 and x_2 are orthogonal, $\langle x_1 x_2 \rangle = 0$ so
$= ||x_1||_2^2 + ||x_2||_2^2$
So both are the same

b)

Proven base case n=2
Inductive hypothesis: true for n orthogonal vecs $\{x_1 \ldots x_n\}$
WTS true for n+1 orthogonal vecs $\{x_1 \ldots x_n \; x_{n+1}\}$
Since we know $||x_1||_2^2 + \ldots + ||x_n||_2^2 = ||x_1 + \ldots x_n||_2^2$,
Let $x = x_1 + \ldots x_n$ so now we need to show $||x + x_{n+1}||_2^2 = ||x||_2^2 + ||x_{n+1}||_2^2$ and since we have this already from part a, we know this is true if x and x_{n+1} are orthogonal
Proving x and x_{n+1} are orthogonal:
Using inner products we know $\langle x, x_{n+1} \rangle = \langle x_1 + \ldots + x_n, x_{n+1} \rangle = \langle x_1, x_{n+1} \rangle + \ldots + \langle x_n, x_{n+1} \rangle$

Since we know that x_{n+1} is orthogonal to $\{x_1 ... x_n\}$, $\langle x_i, x_{n+1}\rangle = 0$
So the sum $= 0$ so x and x_{n+1} are orthogonal

Since they are orthogonal, we can now apply our result from a to complete our inductive step

3

$S^T = -S$

a) WTS I-S is nonsingular

If 0 is an eigenvalue, then a matrix is singular
Proof by contradiction, let 0 be an eigenvalue for I-S
So $\exists x \in \mathbb{R}^m$ where $x \neq 0$ st $(I-S)x = 0$ so $x - Sx = 0$ so $Sx = x$. This implies that 1 is an eigenvalue of S but since we know that S is skew-symmetric, we know that its eigenvalues are imaginary of 0 so 1 cannot be an eigenvalue therefore contradiction.
So 0 is not an eigenvalue for $I-S$ so $I-S$ is nonsingular

b)

WTS $Q = (I-S)^{-1}(I+S)$ is orthogonal ie $Q^T = Q^{-1}$ so

$QQ^T = (I-S)^{-1}(I+S)((I-S)^{-1}(I+S))^T$
$= (I-S)^{-1}(I+S)(I+S)^T(I-S)^{-1T}$
$= (I-S)^{-1}(I+S)(I+S^T)(I-S^T)^{-1}$
$= (I-S)^{-1}(I+S)(I-S)(I+S)^{-1}$
Since $(I+S)(I-S) = I - S^2 = (I-S)(I+S)$ we can commute them so
$= (I-S)^{-1}(I-S) \quad * \quad (I+S)(I+S)^{-1}$
$= I * I = I$

$Q^TQ = ((I-S)^{-1}(I+S))^T(I-S)^{-1}(I+S)$
$= (I+S)^T(I-S)^{-1T}(I-S)^{-1}(I+S)$
$= (I+S^T)(I-S^T)^{-1}(I-S)^{-1}(I+S)$
$= (I-S)(I+S)^{-1}(I-S)^{-1}(I+S)$
Since $(I+S)$ and $(I-S)$ commute, their inverses must compute so $(I+S)^{-1}$ and $(I-S)^{-1}$ commute
$= (I-S)(I-S)^{-1} \quad * \quad (I+S)^{-1}(I+S)$
$= I * I = I$

Therefore $Q^TQ = QQ^T = I$ so $Q^T = Q^{-1}$

4

a)

Rank of $\vec{u}\vec{v}^T$

Let $\vec{u} = (u_1, \dots, u_m)^T \in \mathbb{R}^m$ and $\vec{v} = (v_1, \dots, v_m)^T \in \mathbb{R}^m$. Then:

$$\vec{u}\vec{v}^T = \begin{pmatrix} u_1\vec{v}^T \\ \vdots \\ u_m\vec{v}^T \end{pmatrix} \text{ where } u_i\vec{v}^T = (u_iv_1, \dots, u_iv_m).$$

This means that every row is a multiple of $\vec{v}^T$. So it can be row reduced to a matrix with 1 row filled and the rest 0, which makes the matrix have rank $= 1$.

b)

Contrapositive: $\vec{v}^T\vec{u} = -1 \implies A$ is singular

Let $A = I + \vec{u}\vec{v}^T$.
Assume $\vec{v}^T\vec{u} = -1$ and $\vec{u} \neq \vec{0}$.
$A\vec{u} = (I + \vec{u}\vec{v}^T)\vec{u} = \vec{u} + \vec{u}(\vec{v}^T\vec{u}) = \vec{u} - \vec{u} = \vec{0}$.
Since $A\vec{u} = \vec{0}$ where $\vec{u} \neq \vec{0}$, so A is singular.

c)

Contrapos: A is singular $\implies \vec{v}^T\vec{u} = -1$

Assume A is singular. Then there exists $\vec{x} \neq \vec{0}$ such that $A\vec{x} = \vec{0}$.
$A\vec{x} = \vec{0} \implies (I + \vec{u}\vec{v}^T)\vec{x} = \vec{0} \implies \vec{x} + \vec{u}(\vec{v}^T\vec{x}) = \vec{0} \implies \vec{x} = -(\vec{v}^T\vec{x})\vec{u}$.
Since $(\vec{v}^T\vec{x})$ is a scalar, $\vec{x} = \lambda\vec{u}$ where $\lambda = -\vec{v}^T\vec{x}$.
Since $\vec{x}, \vec{v} \neq \vec{0}$, $\lambda \neq 0$ which means $\vec{u} \neq \vec{0}$ (sanity check)
Substitute $\vec{x} = \lambda\vec{u}$ back into the equation:
$A(\lambda\vec{u}) = \lambda(A\vec{u}) = 0 \implies A\vec{u} = 0$ (since $\lambda \neq 0$).
$(I + \vec{u}\vec{v}^T)\vec{u} = \vec{u} + \vec{u}(\vec{v}^T\vec{u}) = \vec{0}$
$\vec{u}(1 + \vec{v}^T\vec{u}) = \vec{0}$
Since $\vec{u} \neq \vec{0}$, then $1 + \vec{v}^T\vec{u} = 0 \implies \vec{v}^T\vec{u} = -1$.

Let $A^{-1} = I - \frac{1}{1+\vec{v}^T\vec{u}}\vec{u}\vec{v}^T$ WTS $A^{-1}A = I = AA^{-1}$
$(I + \vec{u}\vec{v}^T)(I - \frac{1}{1+\vec{v}^T\vec{u}}\vec{u}\vec{v}^T)$

Arrows omitted for clarity

const const

$$(I + uv^T)(I - \frac{1}{1+v^Tu}uv^T) = I - \frac{1}{1+v^Tu}uv^T + uv^T - \frac{1}{1+v^Tu}u\boxed{v^Tu}v^T =$$

$$I + uv^T - \frac{1}{1+v^Tu}uv^T - \frac{v^Tu}{1+v^Tu}uv^T = I + uv^T - \frac{1+v^Tu}{1+v^Tu}uv^T = I + uv^T - uv^T = I$$

$$(I - \frac{1}{1+v^Tu}uv^T)(I + uv^T) = I + uv^T - \frac{1}{1+v^Tu}uv^T + \frac{1}{1+v^Tu}uv^Tuv^T$$ commutes to be same as abv

5

$\|\cdot\|_W$ is a vector norm

Given $\|\cdot\|$ is a vector norm and $W \in \mathbb{R}^{m\times m}$ is non-singular. Define $\|\vec{x}\|_W = \|W\vec{x}\|$.

1. Positive:
   Since $\vec{x} \in \mathbb{R}^m$ and $W \in \mathbb{R}^{m\times m}$, let $\vec{y} = W\vec{x}$ ($y \in \mathbb{R}^m$).
   $\|\vec{x}\|_W = \|W\vec{x}\| = \|\vec{y}\|$ and since $\|\cdot\|$ is a vector norm $\|\vec{y}\| \geq 0$
   Additionally, since W is non-singular, $W\vec{x} = \vec{0} \iff \vec{x} = \vec{0}$.
   So $\|\vec{x}\|_W = \|W\vec{x}\| = \|\vec{y}\| = 0 \iff \|\vec{y}\| = 0 \iff \vec{x} = \vec{0}$.
2. Addition
   Let $\vec{x}, \vec{y} \in \mathbb{R}^m$.
   $\|\vec{x}+\vec{y}\|_W = \|W(\vec{x}+\vec{y})\| = \|W\vec{x}+W\vec{y}\|$.
   Since $\|\cdot\|$ is a vector norm:
   $\|W\vec{x}+W\vec{y}\| \leq \|W\vec{x}\| + \|W\vec{y}\| = \|\vec{x}\|_W + \|\vec{y}\|_W$.
3. Scalar mult
   Let $\lambda \in \mathbb{R}$.
   $\|\lambda\vec{x}\|_W = \|\lambda W\vec{x}\|$ and since $\|\cdot\|$ is a vector norm:
   $= |\lambda|\|W\vec{x}\| = |\lambda|\|\vec{x}\|_W$.
   $\therefore \|\cdot\|_W$ is a vector norm.

6

a)

$\|\vec{x}\|_\infty \leq \|\vec{x}\|_2$

Let $\vec{x} = (n_1, n_2, \dots, n_m)$. Let $n_j = \max\{n_1, \dots, n_m\}$.

Then:

- $\|\vec{x}\|_\infty = n_j$
- $\|\vec{x}\|_2 = \sqrt{n_1^2 + \cdots + n_j^2 + \cdots + n_m^2}$

Since all elements are squared, all of n_i is positive so, $n_1^2 + \cdots + n_j^2 + \cdots + n_m^2 \geq n_j^2$.
So, $\sqrt{n_1^2 + \cdots + n_m^2} \geq \sqrt{n_j^2} = n_j$.

Therefore:
$\|\vec{x}\|_\infty \leq \|\vec{x}\|_2$

b)

$\|\vec{x}\|_2 \leq \sqrt{m}\|\vec{x}\|_\infty$

Let $\vec{x} = (x_1, \ldots, x_m)$ where $x_j = \max\{x_1, \ldots, x_m\}$.

We know:

- $\|\vec{x}\|_2 = \sqrt{x_1^2 + \cdots + x_m^2} \implies \|\vec{x}\|_2^2 = x_1^2 + \cdots + x_m^2$
- $(\sqrt{m}\|\vec{x}\|_\infty)^2 = (\sqrt{m}x_j)^2 = mx_j^2$

Since $\forall i \in [1..m]$, $x_i \leq x_j$:
$\|\vec{x}\|_2^2 \leq x_j^2 + \cdots + x_j^2$ (m times) $= mx_j^2$

So:
$(\|\vec{x}\|_2)^2 \leq (\sqrt{m}\|\vec{x}\|_\infty)^2 \implies \|\vec{x}\|_2 \leq \sqrt{m}\|\vec{x}\|_\infty$

c)

$\|A\|_\infty \leq \sqrt{n}\|A\|_2$

Let $x \in \mathbb{R}$ therefore Ax is a vector
$||Ax||_\infty \leq ||Ax||_2$ from prev $\leq ||A||_2||x||_2$ by 2-norm rules
$\leq ||A||_2\sqrt{n}||x||_\infty$ by prev

So now we have $||Ax||_\infty \leq ||A||_2\sqrt{n}||x||_\infty$ which implies
$\frac{||Ax||_\infty}{||x||_\infty} \leq \sqrt{n}||A||_2$
and since $||A||_\infty = sup\frac{||Ax||_\infty}{||x||_\infty}$, $||A||_\infty \leq \sqrt{n}||A||_2$

7

a)

```
function func(n)
    T = zeros(n);
    X = zeros(n);

    for i= 1:n;
        for j= 1:n;

            if i==j
                T(i,j) = 4;
            elseif i==j+1
                T(i,j) = 1;
            elseif i+1==j
                T(i,j) = 1;
            end

            X(i,j) = sqrt(2/(n+1)) * sin ((i * j * pi) / (n+1));
        end
    end

    A = X * X;
    B = X * T * X;

    n
    A
    B
    T
    X
end

func(3)
func(5)
func(7)
```

```
n =

     3


A =

    1.0000         0   -0.0000
         0    1.0000    0.0000
   -0.0000    0.0000    1.0000


B =

    5.4142         0   -0.0000
         0    4.0000    0.0000
   -0.0000    0.0000    2.5858


T =

     4     1     0
     1     4     1
     0     1     4


X =

    0.5000    0.7071    0.5000
    0.7071    0.0000   -0.7071
    0.5000   -0.7071    0.5000
```

```
n =

     5


A =

    1.0000    0.0000   -0.0000    0.0000    0.0000
    0.0000    1.0000    0.0000   -0.0000   -0.0000
   -0.0000    0.0000    1.0000   -0.0000    0.0000
    0.0000   -0.0000   -0.0000    1.0000   -0.0000
    0.0000   -0.0000    0.0000   -0.0000    1.0000


B =

    5.7321    0.0000   -0.0000    0.0000    0.0000
    0.0000    5.0000    0.0000   -0.0000   -0.0000
   -0.0000    0.0000    4.0000   -0.0000    0.0000
    0.0000   -0.0000   -0.0000    3.0000   -0.0000
    0.0000   -0.0000    0.0000   -0.0000    2.2679


T =

     4     1     0     0     0
     1     4     1     0     0
     0     1     4     1     0
     0     0     1     4     1
     0     0     0     1     4


X =

    0.2887    0.5000    0.5774    0.5000    0.2887
    0.5000    0.5000    0.0000   -0.5000   -0.5000
    0.5774    0.0000   -0.5774   -0.0000    0.5774
    0.5000   -0.5000   -0.0000    0.5000   -0.5000
    0.2887   -0.5000    0.5774   -0.5000    0.2887
```

```
n =

     7


A =

    1.0000    0.0000   -0.0000         0   -0.0000   -0.0000   -0.0000
    0.0000    1.0000    0.0000   -0.0000    0.0000    0.0000    0.0000
   -0.0000    0.0000    1.0000   -0.0000   -0.0000    0.0000   -0.0000
         0   -0.0000   -0.0000    1.0000    0.0000   -0.0000    0.0000
   -0.0000    0.0000   -0.0000    0.0000    1.0000   -0.0000   -0.0000
   -0.0000    0.0000    0.0000   -0.0000   -0.0000    1.0000    0.0000
   -0.0000    0.0000   -0.0000    0.0000   -0.0000    0.0000    1.0000


B =

    5.8478   -0.0000   -0.0000    0.0000   -0.0000   -0.0000   -0.0000
   -0.0000    5.4142    0.0000   -0.0000    0.0000    0.0000    0.0000
   -0.0000    0.0000    4.7654         0   -0.0000    0.0000   -0.0000
    0.0000   -0.0000   -0.0000    4.0000    0.0000   -0.0000    0.0000
   -0.0000    0.0000   -0.0000    0.0000    3.2346   -0.0000   -0.0000
   -0.0000    0.0000    0.0000   -0.0000   -0.0000    2.5858    0.0000
   -0.0000    0.0000   -0.0000    0.0000   -0.0000    0.0000    2.1522


T =

     4     1     0     0     0     0     0
     1     4     1     0     0     0     0
     0     1     4     1     0     0     0
     0     0     1     4     1     0     0
     0     0     0     1     4     1     0
     0     0     0     0     1     4     1
     0     0     0     0     0     1     4


X =

    0.1913    0.3536    0.4619    0.5000    0.4619    0.3536    0.1913
    0.3536    0.5000    0.3536    0.0000   -0.3536   -0.5000   -0.3536
    0.4619    0.3536   -0.1913   -0.5000   -0.1913    0.3536    0.4619
    0.5000    0.0000   -0.5000   -0.0000    0.5000    0.0000   -0.5000
    0.4619   -0.3536   -0.1913    0.5000   -0.1913   -0.3536    0.4619
    0.3536   -0.5000    0.3536    0.0000   -0.3536    0.5000   -0.3536
    0.1913   -0.3536    0.4619   -0.5000    0.4619   -0.3536    0.1913
```

b)

$X = X^T = X^{-1}$

Since XTX is a diagonalized matrix, this means that XTX has the eigenvalues as its values so X has eigenvectors as its elements